

PROJECT PROPOSAL

Modular "Plug-and-Play" AI-Agent Platform for Enterprise Onboarding

Document Type:	Architecture & Feature Proposal	Prepared For:	Business Stakeholders & Technical Leads
Core Focus:	Zero-Friction Business Onboarding & Automated MCP Tool Generation	Status:	Updated Version

1. Executive Summary

This proposal outlines a modular, plug-and-play architecture for an AI-driven customer support and CRM management platform. The primary goal of this solution is **zero-friction onboarding**: enabling any new business owner to integrate their business in minutes simply by loading user data, domain policies, and knowledge bases.

Upon document upload, the platform automatically ingests domain knowledge into a Retrieval-Augmented Generation (RAG) engine and dynamically auto-generates custom **Model Context Protocol (MCP)** function files. These MCP files enable AI agents to safely execute targeted operational actions directly on tenant-specific user data without requiring manual code writing or manual API configuration.

Key Value Proposition

Eliminates developer overhead during client integration. Business owners simply drop their policies and customer schemas into the system, and the platform automatically creates the functional tools necessary to serve end users and manage customer data.

2. "Plug-and-Play" System Architecture



3. Core Plug-and-Play Capabilities

A. Automated MCP Function Generation Automation Engine

- **Schema Auto-Discovery:** When a business owner uploads customer data schemas, API specifications, or database structures, the system automatically builds structured Model Context Protocol (MCP) tool definitions.
- **Dynamic Function Binding:** The system automatically writes the executable logic needed to perform core CRUD operations (Create, Read, Update, Delete) on tenant data without developer intervention.

B. Zero-Code Domain & Policy Ingestion

- **Instant Knowledge Base Building:** Uploaded business documentation (PDFs, user guides, policy sheets) are parsed and chunked into isolated vector indices dedicated strictly to that tenant.
- **Automated Guardrails & Compliance:** The platform reads policy guidelines and dynamically creates operational guardrails to ensure the AI agent adheres to company-specific rules and security boundaries.

C. Dual-Persona Operational Interfaces

- **Customer Support Agent:** Handles incoming multi-modal requests (text, voice messages, images). It answers queries grounded in tenant policy via RAG and executes transactional actions (e.g., ticket creation, account updates) using auto-generated MCP functions.

- **Business Owner Control Center:** A dedicated administrative chatbot interface that allows owners to update domain policies, audit AI-executed actions, or modify generated tool behaviors using plain natural language.

D. Escalation & Safety Mechanisms

- **Automated Fallback:** If a user request falls outside the auto-generated MCP tool boundaries, contains ambiguous data, or violates business compliance rules, the system seamlessly routes the conversation to a human support agent.

4. Step-by-Step Business Onboarding Workflow

1

Drop & Load Phase: The new business owner accesses the portal and uploads raw policy documents, domain knowledge files, and user database schemas/API specs.

2

Automated Generation Phase:

- The *RAG Ingestion Engine* indexes and vectorizes all knowledge documents into a isolated workspace.
- The *MCP Auto-Generator* parses user schemas and instantly outputs functional code/tool definitions to handle actions on user data.

3

Instant Activation: The AI chatbot goes live instantly—fully equipped with exact domain context, operational boundaries, and dynamic tool capabilities to manipulate customer data safely.